

ECSE-4040

Digital Electronics

Lecture 11

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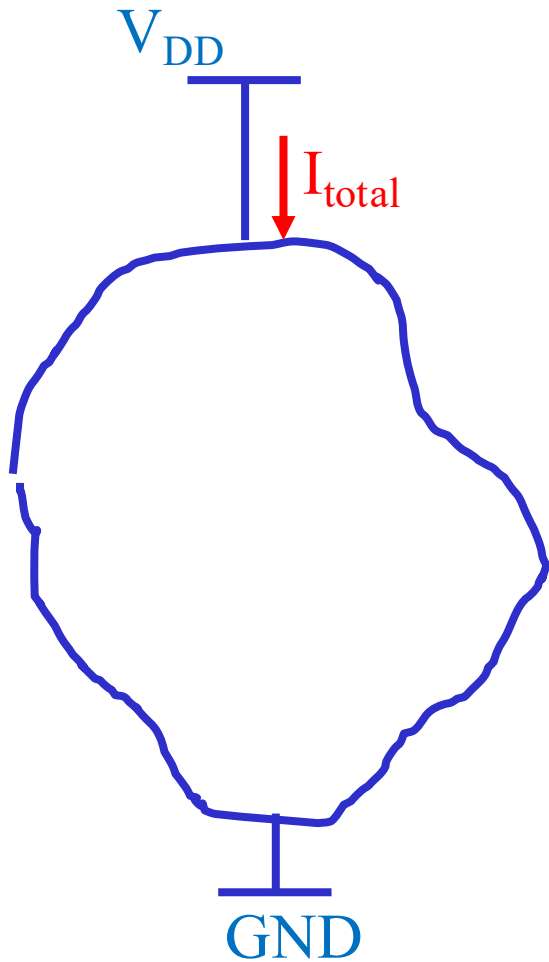
Lecture Outline

- Dynamic and Static Power Consumption
- Power and Delay Tradeoffs

CMOS Power Consumption

- Affects chip temperature, packaging decisions, and long-term reliability.
- Has a static component (steady state) & a dynamic component (when switching from one state to the other)
- To minimize power, we often adjust the timing characteristics of a design
- There is a tradeoff between power and timing → a key issue in CMOS digital design.

Power Consumption



$$P = V_{DD} * I_{total}$$

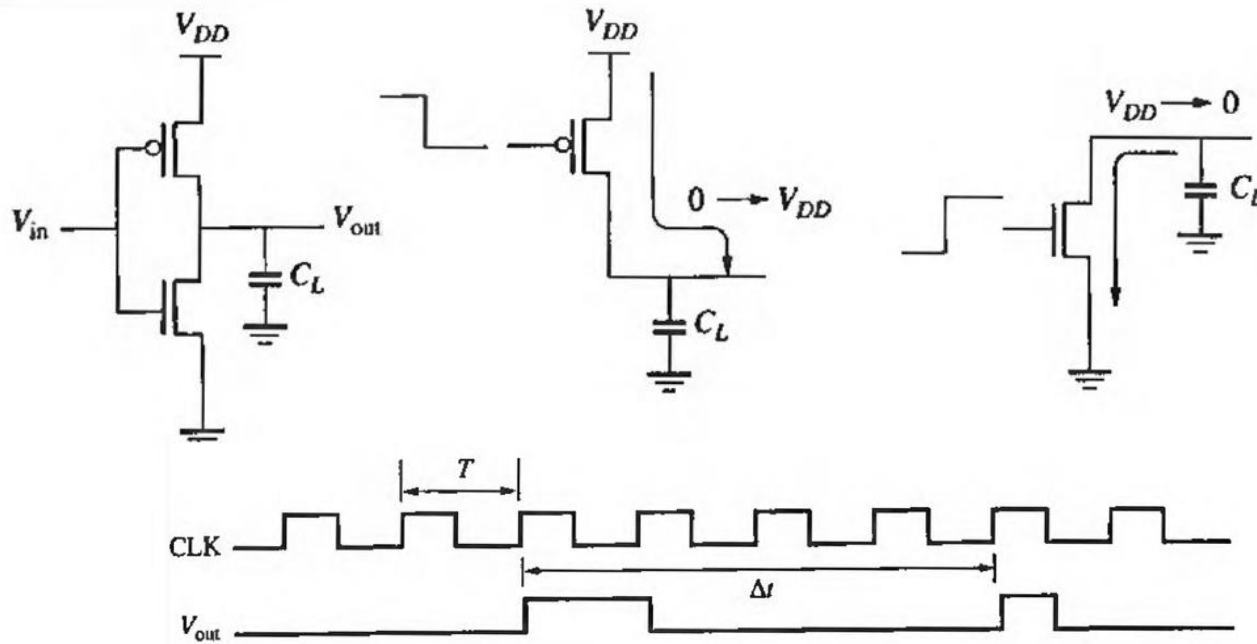
static

- Leakage current
- dc standby current (Only in NMOS/pseudo-nmos)

Dynamic

- Capacitance switching
- Current flowing during switching
“crowbar” current
- Power due to glitches in the output waveform

(Dynamic) Switching Power

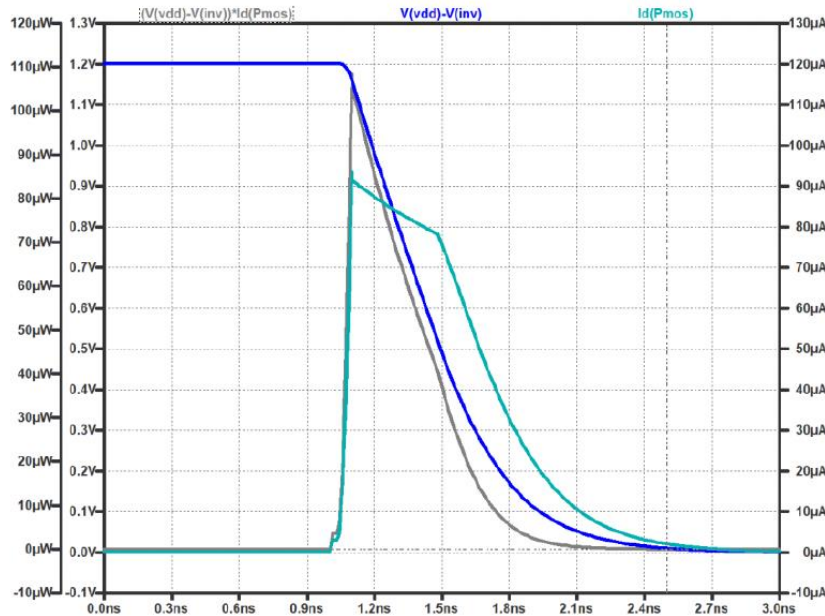


$$P_{switching} = V_{DD} * I_{D,avg}$$

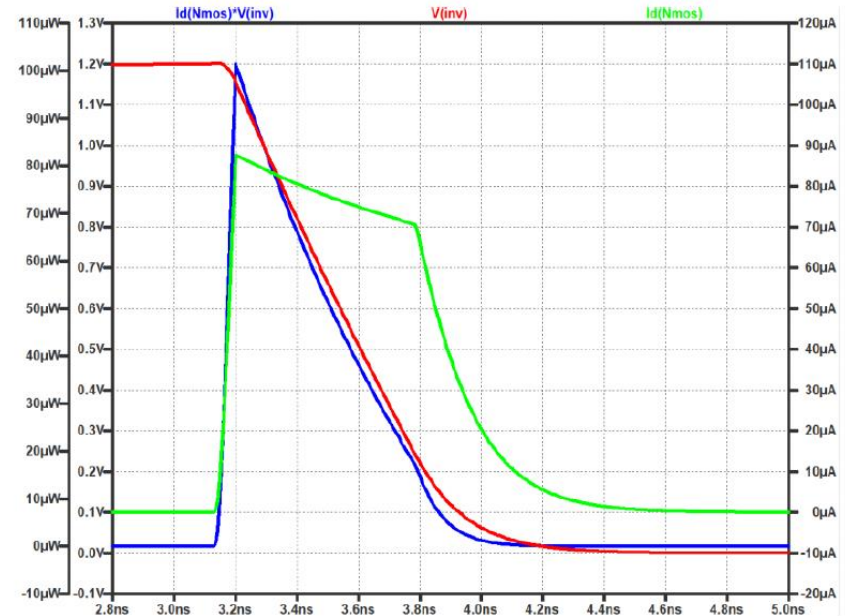
$$I_{D,avg} = C \frac{dV}{dt} = C_L \frac{\Delta V_{swing}}{\Delta t} = C_L V_{DD} f_{avg}$$

$$P_{switching} = C_L V_{DD}^2 f_{avg}$$

(Dynamic) Switching Power



$$P_{charge} = 1/2 C_L V_{DD}^2 f_{avg}$$

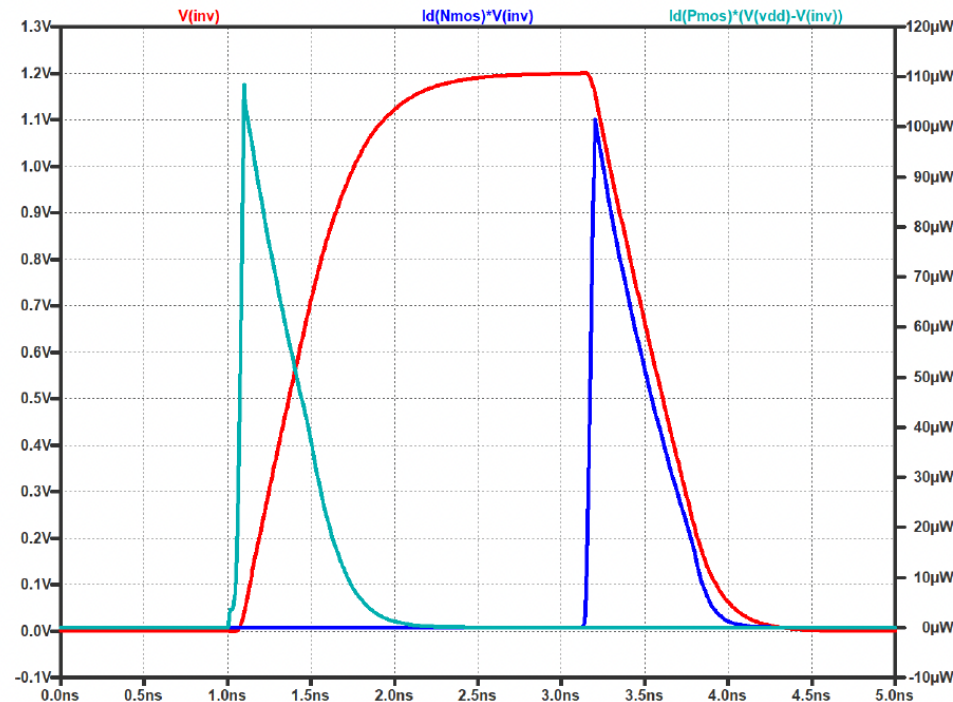


$$P_{discharge} = 1/2 C_L V_{DD}^2 f_{avg}$$

$C_L = 50\text{fF}$
 $L = 100\text{nm}$
 $W_P = 2.4 * L$
 $W_N = 1 * L$

$$P_{switching} = C_L V_{DD}^2 f_{avg}$$

(Dynamic) Switching Power

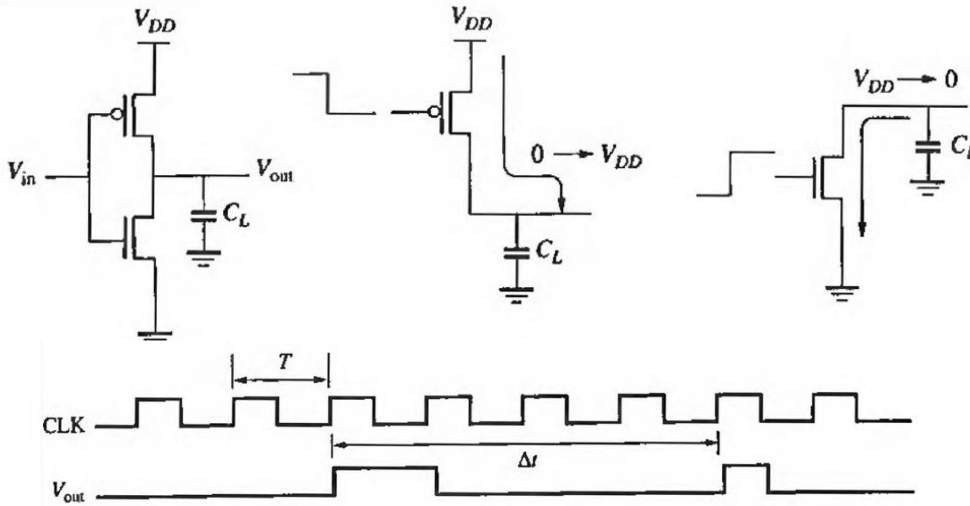


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(Dynamic) Switching Power



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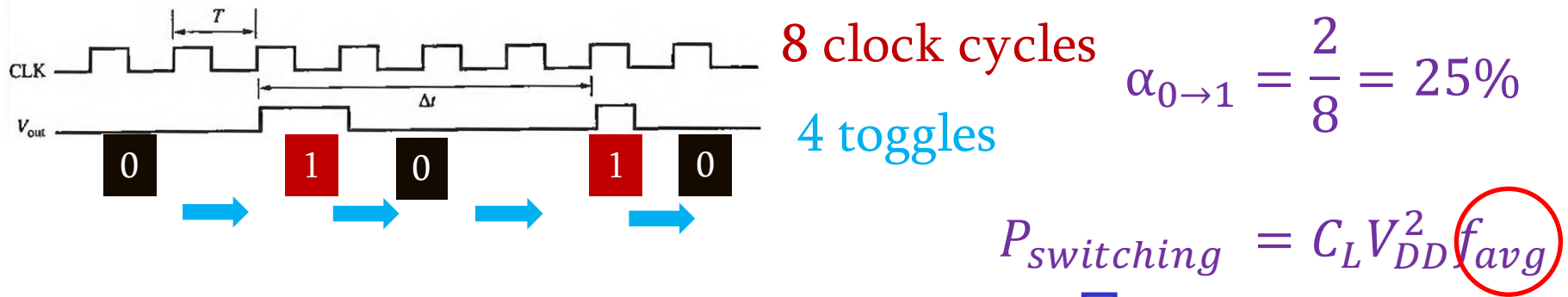


Gate sizing, interconnect capacitances

Reduce $V_{DD} \rightarrow$ noise margins?

Architecture dependent (affects area as well), timing constraints,

(Dynamic) Switching Power



Most logic gates do not switch on every cycle → let us define an activity factor $\alpha_{0 \rightarrow 1}$

$$P_{switching} = \alpha_{0 \rightarrow 1} C_L V_{DD}^2 f_{clk}$$

$$\alpha_{0 \rightarrow 1} = \frac{\# \text{ toggles} / 2}{\# \text{ clock cycles}}$$

(Dynamic) Switching Power Calculations vs. Simulations

Comparison of Hand Calculation of Power with SPICE

Compute the power due to switching for an inverter with $W_p = 800$ nm and $W_n = 400$ nm driving a total load of 50 fF with an average switching frequency of 250 MHz. Compare the results with a transient simulation in SPICE. Which value is larger, the one obtained by hand calculation or by SPICE? Use 0.18 μ m technology parameters for the analysis with $V_{DD} = 1.8$ V. Ignore parasitic capacitances and set input rise/fall times to 300 ps in SPICE.

Solution:

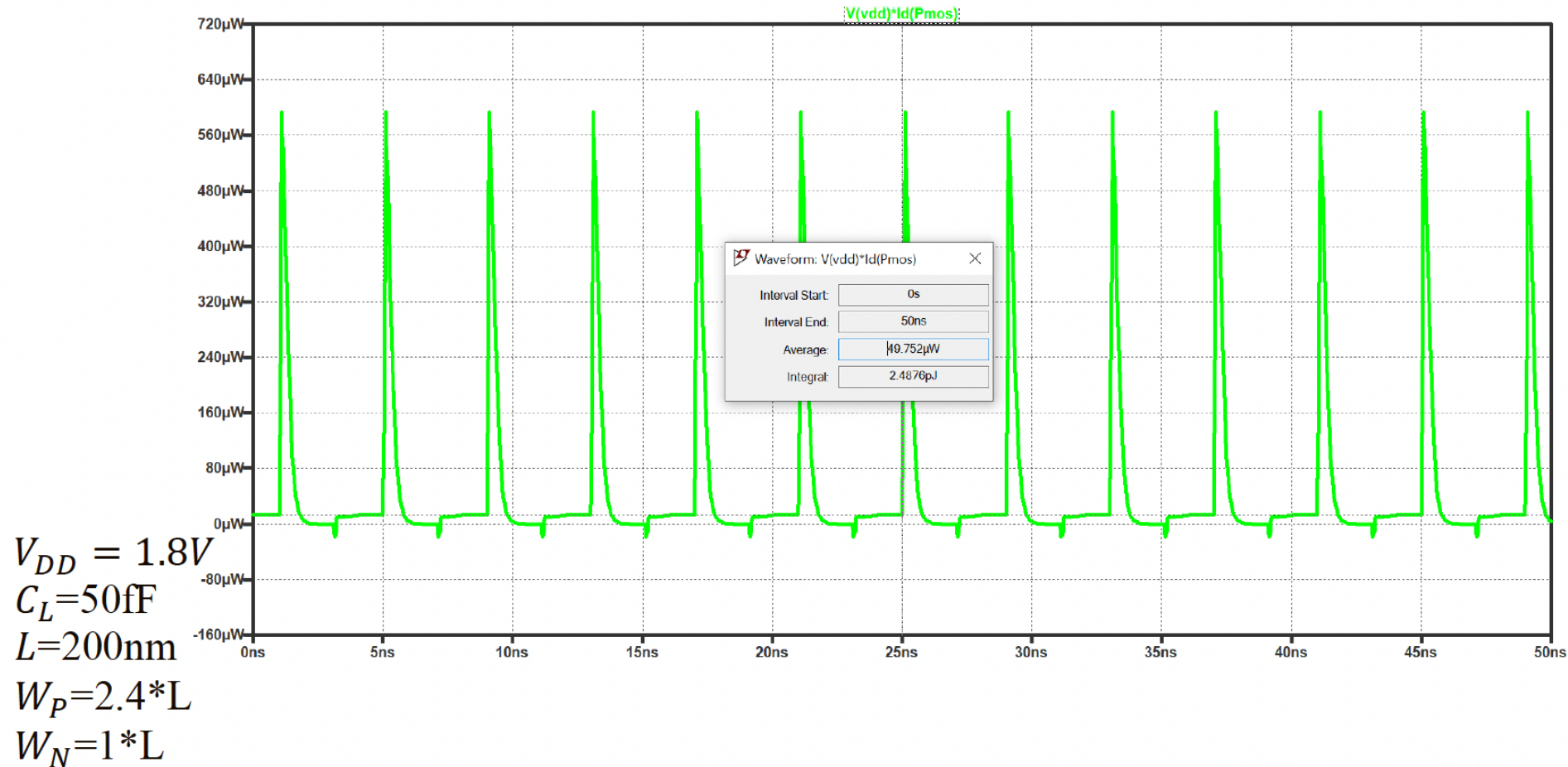
$$P_{SPICE} > C_L V_{DD}^2 f_{clk}$$

The dynamic power can be computed by hand as

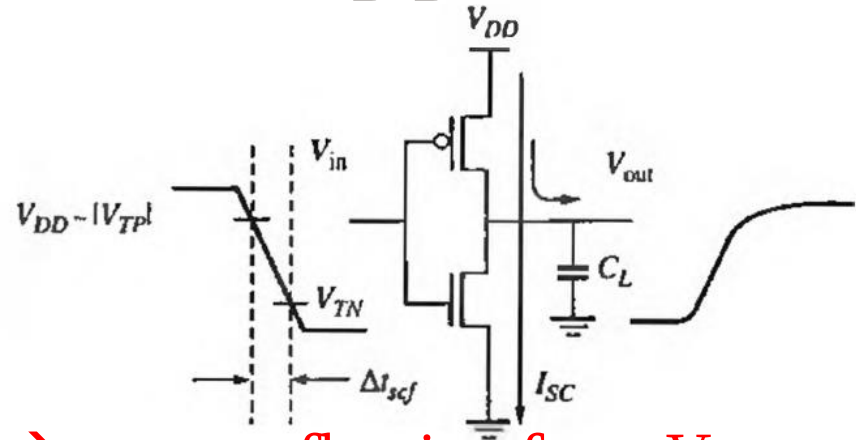
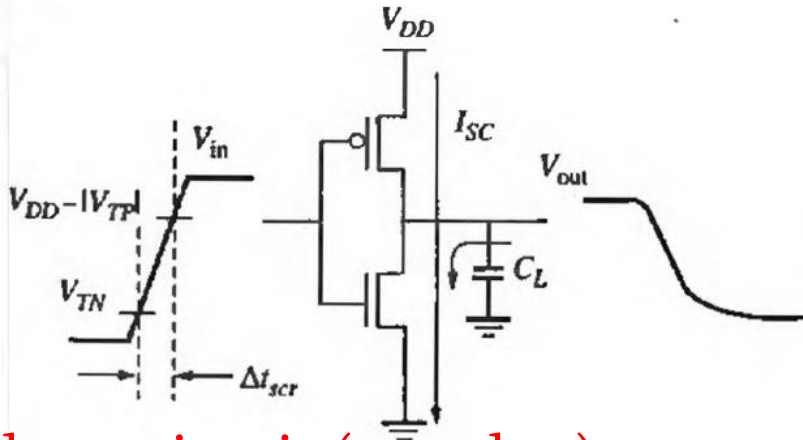
$$P = CV^2f = (50 \text{ fF})(1.8)^2(250 \text{ MHz}) = 40.5 \text{ } \mu\text{W}$$

A SPICE simulation can be performed to confirm this result. A transient analysis is required, followed by an average power measurement over a 4 ns period. Note that the inverter must toggle high and low in order to create a “power event.” The input

(Dynamic) Switching Power Calculations vs. **Simulations**



Why $P_{SPICE} > C_L V_{DD}^2 f_{clk}$



Short-circuit (crowbar) current $\rightarrow$ current flowing from V_{DD} to GND during switching (2 transistors in the path are conducting)

$$\Delta t_{sc} = \Delta t_{scr} + \Delta t_{scf}$$

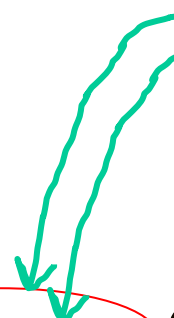
$$P_{SC} = V_{DD} * I_{SC}$$

$$I_{SC} = \frac{\Delta t_{sc}}{T} * I_{SC,avg}$$

$$P_{SC} = V_{DD} * \frac{\Delta t_{sc}}{T} * I_{SC,avg} = V_{DD} \Delta t_{sc} I_{SC,avg} f_{clk}$$

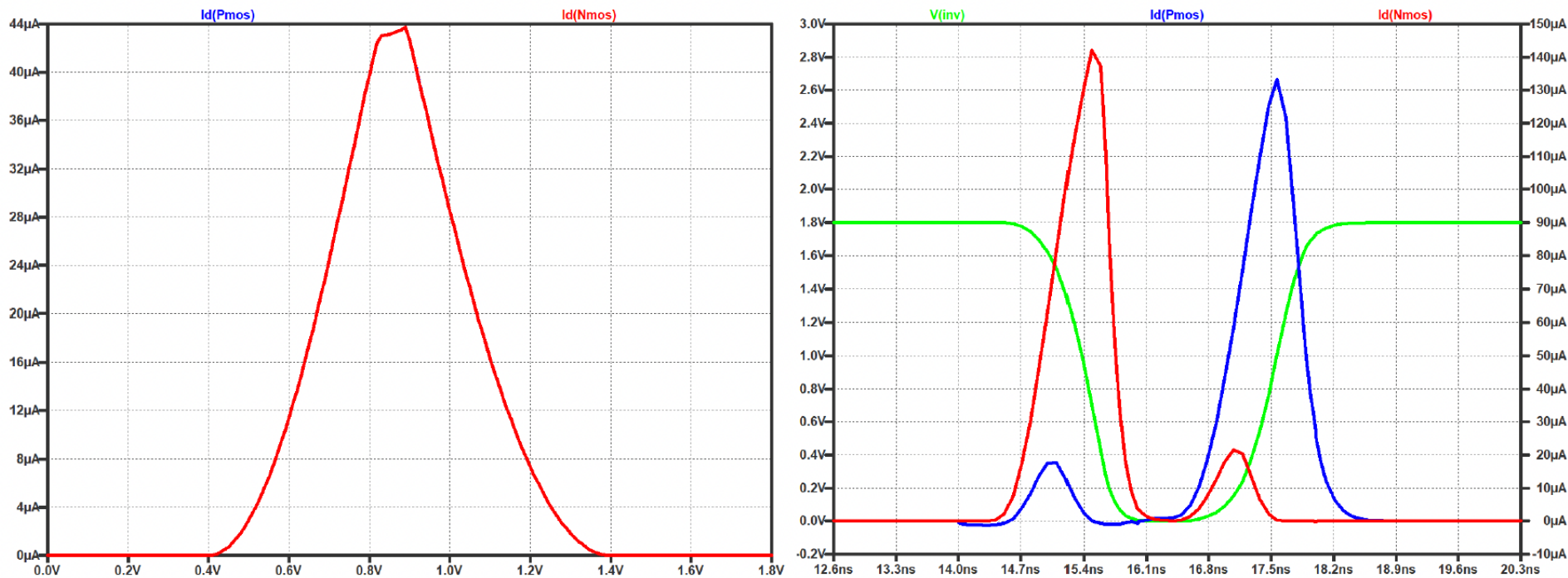
$$P_{SC} = C_{SC} V_{DD}^2 f_{clk}$$

$$I = C \frac{dV}{dt}$$



Why $P_{SPICE} > C_L V_{DD}^2 f_{clk}$

Short-circuit (crowbar) current $\rightarrow$ current flowing from V_{DD} to GND during switching (2 transistors in the path are conducting)



$$P_{SC} = C_{SC} V_{DD}^2 f_{clk}$$

(Dynamic) Switching Power

$$P_{switching} = V_{DD} * I_{D,avg}$$

$$P_{switching} = \alpha_{0 \rightarrow 1} C_L V_{DD}^2 f_{clk} + C_{sc} V_{DD}^2 f_{clk}$$

$$C_{sc} = \alpha_{sc} C_L$$

$$P_{switching} = \alpha C_L V_{DD}^2 f_{clk}$$

α : Both activity factor and short circuit current efforts

To reduce power $\rightarrow$ reduce rise and fall times (edge rates) $\rightarrow$ minimize Δt_{sc} BUT this is local optimization—Not necessarily globally optimal because if you increase devices dimensions to reduce times $\rightarrow$ capacitances will increase

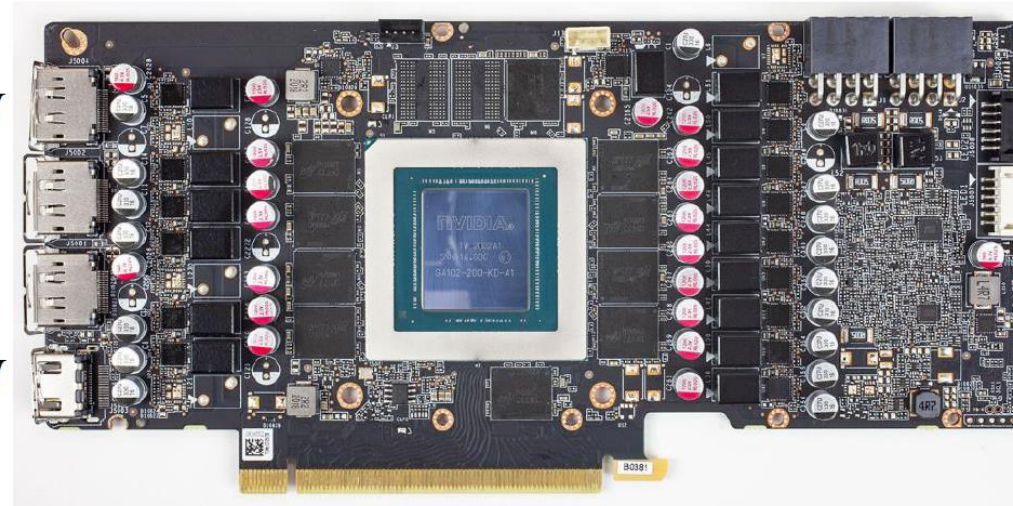
Tradeoff between dynamic power of the previous stage
and short-circuit power of the next gate

(Dynamic) Switching Power

Just how important is α ?

$$P_{switching} = \alpha C_L V_{DD}^2 f_{clk}$$

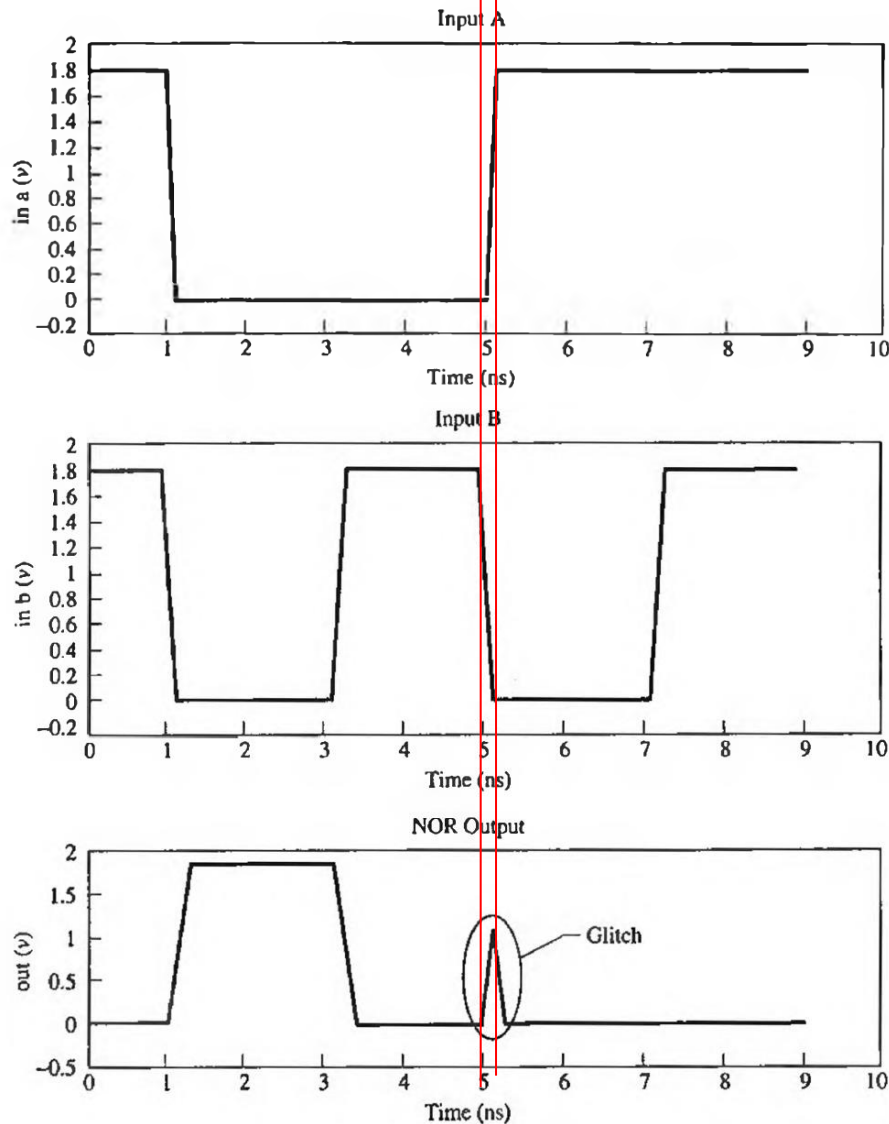
Nvidia RTX 3080



<https://www.techpowerup.com/gpu-specs/nvidia-ga102.g930>

- $V_{DD} = 1\text{ V}$
- $f_{CLK} = 2\text{ GHz}$
- 28.3B transistors
 - $C_L = 0.5\text{ fF}$ per transistor
 - $C_L = 15\text{ }\mu\text{F}$ total
 - If $\alpha = 1$, $P_{switching} = 30\text{ kW}$
 - Alternatively, $I_{avg} = 10\text{ }\mu\text{A}$ per transistor
 - $I_{avg} = 30\text{ kA}$
 - If $\alpha = 1$, $P_{switching} = 30\text{ kW}$
- Actual $P_{switching} = 300\text{ W}$
- Actual $\alpha = 0.01$

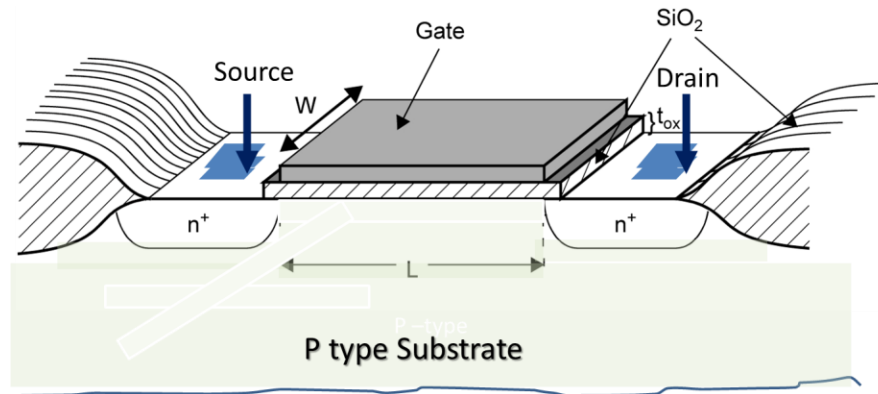
Dynamic Power and Glitches



Difference in the arrival times of the signals at the gate inputs $\rightarrow$ glitches
Glitches tend to propagate and cause unintended transitions in the following stages and increase the power consumption (as well as error in the output)

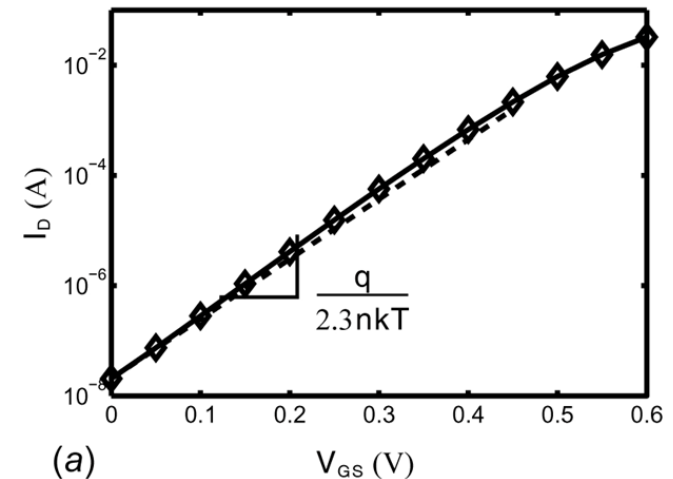
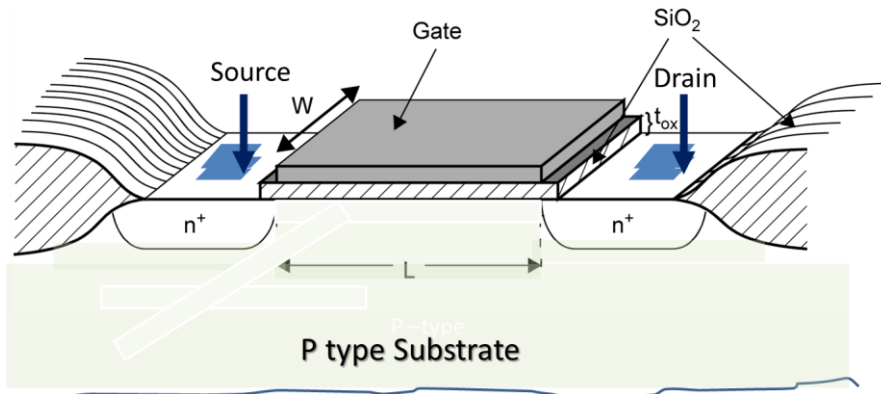
To reduce # of glitches $\rightarrow$ signals should all arrive at the same time at all gates – Balance path delays and gate delays

Static (Standby) Power



- Before the channel is created (strong inversion), there is “tiny” current flowing (sub-threshold current) + there is also leakage from the reverse p-n junction (drain-substrate) (Both CMOS & NMOS)
- For NMOS/Pseudo-NMOS gates $\rightarrow$ current is flowing if input is high and output is low

Static (Standby) Power— Subthreshold leakage

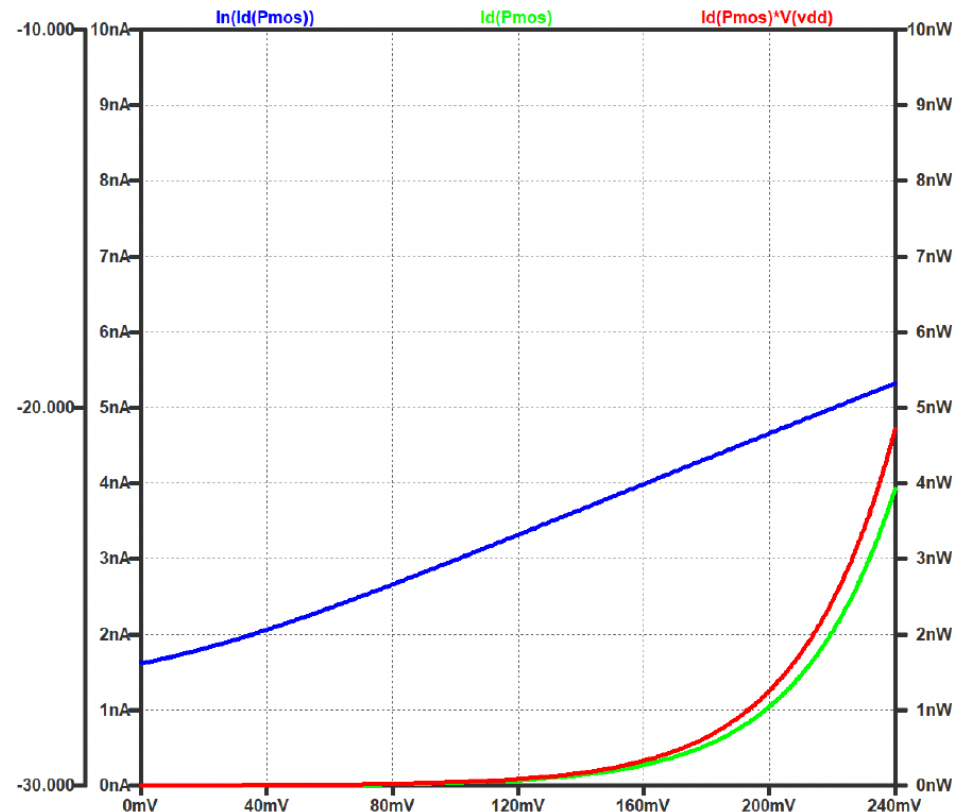


For $V_{GS} < 100\text{mV}$, the channel is not yet complete and there is no drift current component.

The main current component is diffusion component, which is modeled by an exponential relation (similar to BJT)

$$I_{\text{sub}} = I_s e^{[q(V_{GS} - V_T - V_{\text{offset}})/nkT]} (1 - e^{(-qV_{DS}/kT)})$$

Static (Standby) Power— Subthreshold leakage



$$I_{sub} = I_s e^{[q(V_{GS} - V_T - V_{off})/nkT]} (1 - e^{-qV_{DS}/kT})$$

Total Power Dissipation

$$P = P_{switching} + P_{static}$$

$$P = P_{switching} + V_{DD} * (I_{sub} + I_{pn-leakage})$$

$$P_{switching} = \alpha C_L V_{DD}^2 f_{clk}$$
$$I_{pn-leakage} = A_j * I_s \left(e^{\frac{q}{KT} * V_j} - 1 \right)$$

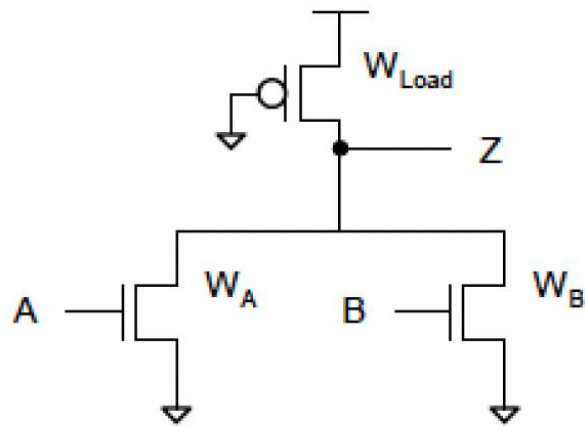
CMOS

$$P = P_{switching} + P_{static} + \frac{I_{ON} * V_{DD}}{2}$$

Pseudo-NMOS

RPI

Static Power in Pseudo-NMOS Gates



How do calculate the static power consumption when given (V_{OL} , V_{OH})?

$$I_{DC} = \frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{OL}}{E_{CN} L_N}\right)} \left[(V_{DD} - V_{TN})(V_{OL}) - \frac{(V_{OL})^2}{2} \right]$$

Power and Delay Tradeoffs

- Equalizing input arrival times reduces glitches
- Equalizing edge rates should minimize short-circuit current.
- All this has to be done while considering power consumption
- We can't reduce power consumption to the point where it starts slowing down the system → We need to define a FOM

$$PDP = P_{avg} * t_p$$

$$PDP \approx CV_{DD}^2 f_{clk} \frac{1}{2f_{CLK}} \approx \frac{1}{2} CV_{DD}^2$$

Energy required to perform a specific operation

Energy-Delay Product

- PDP doesn't capture all design compromises
- PDP can be reduced by lowering $C \rightarrow$ smaller transistors $\rightarrow$ smaller transistors have lower handling capability $\rightarrow$ more rise/fall times and more delays

$$EDP = PDP * t_p$$

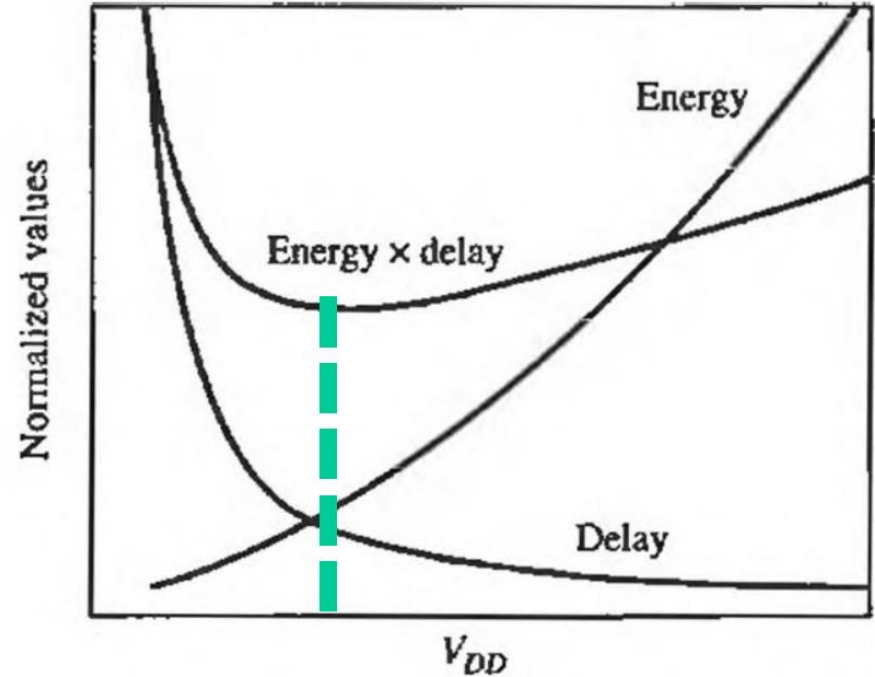
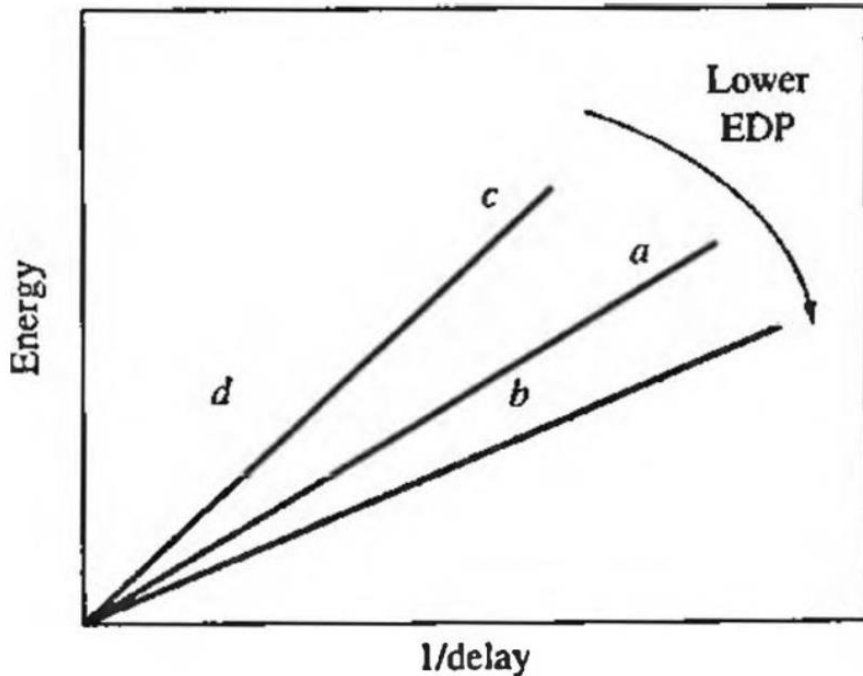
$$I = C \frac{dV}{dt} \rightarrow \Delta t = C \frac{\Delta V}{I}$$

$$t_p = C \frac{\Delta V}{I_{sat}} \approx \frac{C \Delta V}{2K_2(V_{DD} - V_T)}$$

$$EDP \approx \frac{C^2 V_{DD}^3}{2K_2(V_{DD} - V_T)}$$

K_2 : constant (depends on device sizes)

Energy-Delay Product



$$EDP \approx \frac{C^2 V_{DD}^3}{2K_2(V_{DD} - V_T)} \rightarrow \frac{\delta EDP}{\delta V_{DD}} = 0 \rightarrow V_{DD} = \frac{3}{2} V_T$$

Example

Compare the chip power for two cases. In one case, a chip has 10M gates, an activity factor of 10%, $V_{DD} = 1.8$ V, a clock frequency of 500 MHz and an average capacitance per node of 20 fF. In the second case, a chip has 50M gates, an activity factor of 5% $\gg V_{DD} = 1.2$ V, a clock frequency of 1 GHz and an average capacitance per node of 10 fF. Which design is better and why?

Case 1: 10M gates, $a = 0.1$, $C = 20$ fF, $V_{DD} = 1.8$ V, $f_{clk} = 500$ MHz $\rightarrow P = 32.4$ W

Case 2: 50M gates, $a = 0.05$, $C = 10$ fF, $V_{DD} = 1.2$ V, $f_{clk} = 1$ GHz $\rightarrow P = 36$ W

The two power levels are about the same, so it appears that the two designs are about equivalent.

Now compute the energy-delay product.

Case 1: $EDP = PDP * t_p = 32.4 (2 \text{ ns})^2 = 130(10^{-18})\text{J-s}$

Case 2: $EDP = PDP * t_p = 36 (1 \text{ ns})^2 = 36(10^{-18})\text{J-s}$

The second design has a smaller energy-delay product and therefore is superior.